Shusen Ni

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Ph.D. Candidate in Electrical Engineering, Tsinghua University, Beijing, China

Research Interests: high-performance control and fault-tolerant control of multiphase motor drives.

EDUCATION

2022 – 2027 **Ph.D., Electrical Engineering, Tsinghua University**
Department of Electrical Engineering
Advisor: Prof. Zedong Zheng **GPA:** 3.9/4.0
Selected Honors: **2025 China Association for Science and Technology (CAST) Young Science and Technology Talent Cultivation Program (Doctoral Student Special Plan)**; Outstanding Student Scholarship (Top 5%), three times

2018 – 2022 **B.Eng., Electrical Engineering and Automation, Shandong University**
Department of Electrical Engineering
GPA: 94.2/100 **Rank:** 1/241
Selected Honors: **Honorary Bachelor's Degree (Top 2%)**; **President Scholarship (Top 0.1%)**; National Scholarship (Top 2%), three times; Outstanding Graduate

RESEARCH PUBLICATIONS

Journal Papers

J1 **S. Ni**, L. Xu, C. Li, and Z. Zheng, "Fundamental Current Decoupling of Multiphase Machines Under Asymmetric Operation Using Generalized Symmetrical Components and Virtual Phase Shifting," in ***IEEE Transactions on Power Electronics***, doi: 10.1109/TPEL.2026.3723847. [link]

J2 **S. Ni**, L. Xu, C. Li, and Z. Zheng, "Topology and Voltage-Balancing Control of Novel Asymmetrical Six-Phase OEW-PMSM Drives Toward Higher DC-Link Voltage Applications," ***IEEE Transactions on Power Electronics***, vol. 41, no. 3, pp. 3182–3186, Mar. 2026, doi: 10.1109/TPEL.2025.3621744. [link]

J3 **S. Ni**, C. Li, and Z. Zheng, "An Online Optimal Phase Current Derating Method With Arbitrary Current Limits for Multiphase Motors in Subnormal Health," ***IEEE Transactions on Industrial Electronics***, vol. 72, no. 5, pp. 4651–4660, May 2025, doi: 10.1109/TIE.2024.3482003. [link]

J4 **S. Ni**, C. Li, and Z. Zheng, "Control Strategy of a Hybrid SiC-Si Traction Inverter for Direct-Drive Multiphase PMSMs in Marine Propulsion," ***IEEE Transactions on Power Electronics***, vol. 39, no. 12, pp. 16400–16414, Dec. 2024, doi: 10.1109/TPEL.2024.3434703. [link]

Manuscripts under review

M1 **S. Ni** et al., "Stability Analysis and Controller Design of Novel Asymmetrical Six-Phase OEW-Motor Drive for MVDC Applications," submitted to *IEEE Transactions on Power Electronics*. Under review (major revision).

Conference Papers

C1 **S. Ni**, Z. Zheng, K. Liu, and L. Xu, "Stability Analysis and Voltage Controller Design for Novel Series-Connected DC-Link Topology for MVDC Multiphase Drives," *2025 IEEE 3rd International Power Electronics and Application Symposium (PEAS)*, Shenzhen, China, 2025, pp. 2056–2060, doi: 10.1109/PEAS66638.2025.11403479. **Best Presentation Award**. [link]

C2 **S. Ni**, C. Liu, and Z. Zheng, "Online Capacitor Monitoring Strategy in 3L-NPC Inverter Driven Open Winding Multiphase Drives," *2024 27th International Conference on Electrical Machines and Systems (ICEMS)*, Fukuoka, Japan, 2024, pp. 3581–3585, doi: 10.23919/ICEMS60997.2024.10921492. **Student Paper Award**. [link]

C3 **S. Ni**, Y. Chen, C. Liu, and Z. Zheng, "Online Detection of Resistance in Each Phase of Multi-Sector PMSM for Condition Monitoring," *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, Phoenix, AZ, USA, 2024, pp. 6467–6471, doi: 10.1109/ECCE55643.2024.10860858. [link]

C4 **S. Ni**, L. Peng, and Z. Zheng, "Switching Frequency Optimization Method and Control Strategy of Propulsion Inverter for Multiphase Motors With Hybrid SiC-MOS and Si-IGBT Bridge Legs," *2023 26th International Conference on Electrical Machines and Systems (ICEMS)*, Zhuhai, China, 2023, pp. 2931–2937, doi: 10.1109/ICEMS59686.2023.10345128. **Best Paper Award**. [link]

C5 **S. Ni**, L. Peng, and Z. Zheng, "An Online Phase Current Derating Method With Asymmetrical Limits of Multiphase Motors," *IECON 2023 – 49th Annual Conference of the IEEE Industrial Electronics Society*, Singapore, 2023, pp. 1–6, doi: 10.1109/IECON51785.2023.10312261. [link]

C6 **S. Ni**, Z. Zheng, L. Peng, and Y. Li, "A Hybrid PI-FOC and CCS-MPC Method for Multiple Harmonic Current Suppression in Multiphase Machines," *2023 IEEE International Conference on Predictive Control of Electrical Drives and Power Electronics (PRECEDE)*, Wuhan, China, 2023, pp. 1–7, doi: 10.1109/PRECEDE57319.2023.10174487. [link]

C7 **S. Ni**, Z. Zheng, and J. Sun, "An Online Global Fault-Tolerant Control Strategy for Asymmetrical Multiphase Machines With Minimum Losses in Full Torque Operation Range," *2022 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC Asia-Pacific)*, Haining, China, 2022, pp. 1–6, doi: 10.1109/ITECAsia-Pacific56316.2022.9942120. [link]

C8 X. Sun, Y. Chuan, Y. Wang, S. Li, **S. Ni**, and Z. Zheng, "Comparative Performance Analysis of Fractional-Slot Nine-Phase and Three-Phase Motors Based on Winding Reconfiguration," *2026 IEEE 9th International Electrical and Energy Conference (CIEEC)*, Tianjin, China, 2026, pp. 2001–2008, doi: 10.1109/CIEEC69682.2026.11571994. [link]

C9 Y. Li, **S. Ni**, H. Ma, and Z. Zheng, "Current Harmonic Suppression of Dual Three-Phase PMSM Based on an H_∞ and MPC Hybrid Control," *2025 IEEE International Conference on Predictive Control of Electrical Drives and Power Electronics (PRECEDE)*, Nanjing, China, 2025, pp. 1–6, doi: 10.1109/PRECEDE63178.2025.11130983. [link]

C10 Y. Chen, C. Li, **S. Ni**, and Z. Zheng, "Novel Cell-Level Insights into Third Quadrant Current Paths of Power SiC MOSFET," *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, Phoenix, AZ, USA, 2024, pp. 6754–6758, doi: 10.1109/ECCE55643.2024.10861632. [link]

RESEARCH PROJECTS

Sep. 2022 – Dec. 2025

National Natural Science Foundation of China Key Joint Fund Project
High-power, high-performance multiphase permanent-magnet direct-drive electric propulsion systems
Core Researcher

- Proposed a **fundamental–harmonic current decoupling** method based on generalized symmetrical components and virtual phase shifting for multiphase machines under asymmetric operation.
- Developed a hybrid SiC/Si bridge-leg inverter topology and coordinated control strategy, reducing propulsion-system loss by **30%** and inverter cost by **35%**.
- Proposed a generalized construction method for **series-connected DC-link-capacitor topologies** in multiphase MVDC propulsion systems and developed a coordinated-control-based capacitor-voltage-balancing strategy.
- Developed an online per-phase resistance monitoring method and active phase-current derating strategies for **intelligent fault-tolerant control** of multiphase motor drives.

Jan. 2023 – Dec. 2024

National Key Research and Development Program of China
High-quality intelligent coordinated control systems for in-wheel motors
Core Researcher

- Led **full-cycle controller algorithm development and validation**, spanning prototype development, bench testing, and vehicle-level integration.
- Designed and implemented high-speed flux-weakening control, variable-switching-frequency operation, and control-delay compensation for carrier-phase-shifted PWM.
- Successfully deployed and validated the control system on five prototype vehicles and obtained **vehicle type approval** from China's Ministry of Industry and Information Technology (MIIT).

Sep. 2025 – Present

Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China
Aviation electrical equipment and next-generation megawatt-class generation systems
Core Researcher

- Contributed to project proposal development and system-architecture planning for next-generation megawatt-class aviation electrical generation systems.
- Conducted preliminary research on **SiC-based megawatt-class** power-conversion solutions for aviation electrical generation systems.

Sep. 2025 – Apr. 2026

Chenxiang Innovation Award Project, CRRC Zhuzhou Institute Co., Ltd.
Equivalent-circuit modeling and parameter tuning for high-power high-frequency transformers
Core Researcher

- Developed high-frequency transformer models with parasitic capacitance, leakage inductance, and AC resistance; achieved frequency-analysis error within 1%.
- Contributed to automated software for parameter extraction, modeling, and resonance-frequency analysis.
- Received the **only Gold Award among regular projects**.

TEACHING EXPERIENCE

2023 – 2026 | **Teaching Assistant** | **Electric Machinery**
Tsinghua University
Supported lectures, homework guidance, and experiment supervision. Led **hands-on transformer projects** in which students applied electrical-machine theory to design, build, and test their own transformers.

2023 – 2025 | **Teaching Assistant** | **Electric Machine Design, Analysis, and Control**
Tsinghua University
Helped develop the course and guided undergraduates through the design, fabrication, and testing of a **permanent-magnet synchronous motor**. Guided undergraduates in extending their course projects into research, resulting in a published international conference paper.

WORK EXPERIENCE

Jul.–Sep. 2024

Internship, Dongfeng Motor Corporation R&D Institute
Electric Drive Systems, New Energy Vehicles

- Participated in a motor temperature-estimation project for electric vehicle drive systems; identified the root cause of high-speed motor temperature-estimation bias.

- Proposed and validated corrective solutions that improved online temperature-estimation accuracy by **more than 50%** and supported final **vehicle-level implementation**.

MENTORING

Assisted my advisor in mentoring several students.

Xiaohan Sun, Yicuan Chuan, Yufei Wang, and Sui Li Sep. 2024–Dec. 2024
Undergraduate students, Department of Electrical Engineering, Tsinghua University

Topic: Comparative performance analysis of fractional-slot nine-phase and three-phase motors based on winding reconfiguration.

Publication: X. Sun, Y. Chuan, Y. Wang, S. Li, **S. Ni**, and Z. Zheng, "Comparative Performance Analysis of Fractional-Slot Nine-Phase and Three-Phase Motors Based on Winding Reconfiguration," *2026 IEEE 9th International Electrical and Energy Conference (CIEEC)*, Tianjin, China, 2026, pp. 2001–2008, doi: 10.1109/CIEEC69682.2026.11571994.

Yongbo Li 2023–2025
Joint Master's student, Beijing Institute of Technology (Advisor: Prof. Hongwei Ma)

Topic: High-performance multiphase motor control algorithm based on deep reinforcement learning.

Publication: Y. Li, **S. Ni**, H. Ma, and Z. Zheng, "Current Harmonic Suppression of Dual Three-Phase PMSM Based on an H_∞ and MPC Hybrid Control," *2025 IEEE International Conference on Predictive Control of Electrical Drives and Power Electronics (PRECEDE)*, Nanjing, China, 2025, pp. 1–6, doi: 10.1109/PRECEDE63178.2025.11130983.

Huiping Li 2022–2023
Joint Master's student, School of Electrical and Control Engineering, Liaoning Technical University (Advisor: Prof. Zhiling Ren)

Topic: High-performance multiphase inverter control.

AWARDS AND RECOGNITIONS

Oct. 2026	**National Scholarship for Doctoral Students** Awarded by the Chinese Ministry of Education
2026	**Best Innovation Award, 2026 Young Professionals Exhibition and Competition** Awarded by IET Hong Kong
2026	**Gold Award, CRRC Zhuzhou Institute Chenxiang Innovation Award** Awarded by the Zhuzhou Institute of China Railway Rolling Stock Corporation (CRRC); the only Gold Award among regular projects.
2025	**2025 China Association for Science and Technology (CAST) Young Science and Technology Talent Cultivation Program (Doctoral Student Special Plan)** One of eight nationwide nominees recommended by the China Power Supply Society (CPSS)
2025	**Tsinghua Friends–TBEA Talent Scholarship** Awarded by Department of Electrical Engineering, Tsinghua University
2025	**Outstanding Student Scholarship (Top 5%)** Awarded by Tsinghua University
2025	**Best Presentation Award, IEEE PEAS 2025** Awarded by IEEE and China Power Supply Society
2024	**Outstanding Student Scholarship (Top 5%)** Awarded by Tsinghua University

2024	**Student Paper Award, ICEMS 2024** Awarded at the 27th International Conference on Electrical Machines and Systems
2024	**Second Prize, ABB Cup Intelligent Technology Innovation Competition** Awarded by ABB and the Chinese Association of Automation
2023	**Tsinghua Friends–Orient Cable Talent Scholarship** Awarded by Department of Electrical Engineering, Tsinghua University
2023	**Outstanding Student Scholarship (Top 5%)** Awarded by Tsinghua University
2023	**Best Paper Award, ICEMS 2023** Awarded at the 26th International Conference on Electrical Machines and Systems
2022	**Outstanding Graduate and Outstanding Undergraduate Thesis** Awarded by Shandong University
2022	**Honorary Bachelor's Degree (Top 2%)** Awarded by Shandong University
2021	**President Scholarship (Top 0.1%)** Awarded by Shandong University
2021	**National Scholarship (Top 2%)** Awarded by the Chinese Ministry of Education
2020	**First Prize, National Undergraduate Electronics Design Contest** Awarded by the Department of Education of Shandong Province, China
2020	**Outstanding Student of Shandong Province** Awarded by Shandong Provincial Department of Education
2020	**National Scholarship (Top 2%)** Awarded by the Chinese Ministry of Education
2019	**National Scholarship (Top 2%)** Awarded by the Chinese Ministry of Education
2019	**First Prize, Chinese Mathematics Competitions for College Students** Awarded by the Chinese Mathematical Society

ACADEMIC SERVICE

2025	**Session Chair**, Tsinghua–IET Academic Conference
2024 – Present	**Reviewer**, *IEEE Transactions on Power Electronics*
2025 – Present	**Reviewer**, *IEEE Transactions on Industrial Electronics*
2026 – Present	**Reviewer**, *IEEE Journal of Emerging and Selected Topics in Power Electronics*
2026	**Reviewer**, IEEE Transportation Electrification Conference and Expo (ITEC)
2025 – 2026	**Reviewer**, IEEE Applied Power Electronics Conference and Exposition (APEC)
2024 – 2025	**Reviewer**, IEEE Energy Conversion Congress and Exposition (ECCE)

TECHNICAL SKILLS

Programming Languages: VHDL, Verilog, Python, C#, C/C++

Software: Simplorer, MATLAB/Simulink, Code Composer Studio (CCS), Quartus, Altium Designer, Ansys Maxwell